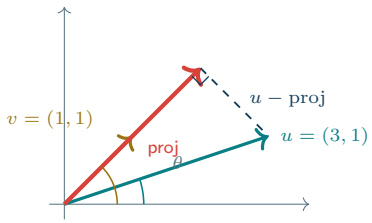
LINEAR ALGEBRA LADDER 2 OF 9

Dot products, length and projection

The dot product measures how far two vectors point the same way, and projection keeps only that part.

1 / THE RULE AND THE PICTURE

The dot product $u \cdot v$ multiplies matching entries and equals $|u||v| \cos\theta$



For $u = (3,1)$ and $v = (1,1)$ we have $u \cdot v = 3 \cdot 1 + 1 \cdot 1 = 4$, while $|u| = \sqrt{10}$ and $|v| = \sqrt{2}$, so

$$\cos\theta = \frac{u \cdot v}{|u||v|} = \frac{4}{\sqrt{10} \cdot \sqrt{2}} = 0.894$$

which gives $\theta = 26.6°$, matching the directions $18.4°$ and $45°$ of the two vectors.

2 / PROJECTION / THE PART OF u ALONG v

Dropping a perpendicular from u onto the line of v gives the projection

$$\text{proj}_v u = \frac{u \cdot v}{v \cdot v} v = \frac{4}{2}(1,1) = (2,2)$$

The remainder $u - \text{proj}_v u = (1,-1)$ is perpendicular to v, since $(1,-1) \cdot (1,1) = 0$, so u splits into a part along v and a part across it.

The length of the projection is $|u| \cos\theta = \sqrt{10} \cdot 0.894 = 2.83 = 2\sqrt{2}$, which is the length of $(2,2)$, and a second case $u = (4,0)$ onto $v = (1,1)$ gives $\frac{4}{2}(1,1) = (2,2)$ as well, with the right-angled triangle of sides $2\sqrt{2}$, $2\sqrt{2}$, 4.

3 / ORTHOGONALITY AND THE RULES

Perpendicular vectors have dot product zero, and lengths obey Pythagoras

rule	statement	example
perpendicular	$u \cdot v = 0$	$(1,2) \cdot (2,-1) = 0$
Pythagoras	$\lvert u+v\rvert^2 = \lvert u\rvert^2 + \lvert v\rvert^2 + 2u \cdot v$	$\lvert(4,2)\rvert^2 = 20 = 10 + 2 + 8$
Cauchy and Schwarz	$\lvert u \cdot v\rvert \le \lvert u\rvert\lvert v\rvert$	$4 \le \sqrt{10} \cdot \sqrt{2} = 4.47$
triangle inequality	$\lvert u+v\rvert \le \lvert u\rvert + \lvert v\rvert$	$4.47 \le 3.16 + 1.41 = 4.57$

For perpendicular vectors the cross term $2u \cdot v$ vanishes, which is Pythagoras' theorem, and $\cos\theta \le 1$ is the Cauchy and Schwarz inequality.

4 / THE PROJECTION MATRIX / AND THE NORMAL OF A LINE

Projection onto v is a matrix, and the line $ax + by = c$ has normal (a,b)

Writing the projection as $P = \dfrac{vv^T}{v^T v} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ for $v = (1,1)$, we have $Pu = (2,2)$ for $u = (3,1)$ and $P^2 = P$, since projecting twice changes nothing.

The equation $ax + by = c$ says $(a,b) \cdot (x,y) = c$, so every point on the line has the same component $c/|(a,b)|$ along the normal (a,b), which is why that vector is perpendicular to the line and why the distance from the origin is $c/\sqrt{a^2+b^2}$.

ASIDE / WORK AND ANGLES

Work is force dot displacement, $W = F \cdot d$, so a force perpendicular to the motion does none.

LINKS TO OTHER SHEETS

The normal vector is developed in *The line $ax + by = c$*; the previous sheet is *Vectors, combinations and the two pictures*, and the next is *Matrices as transformations*.

Convention. $u \cdot v = \sum u_i v_i$; $|u| = \sqrt{u \cdot u}$; θ is the angle between u and v.